The Himalayas

- These are young fold mountains of tertiary period which were folded over Tethys Sea due to inter-continental collision.

- These extend for about 2500 km between Indus and Brahmaputra gorges from west to east in an arc shaped manner.

- The upliftment of Himalayas is believed to have taken place in three main phases after upliftment of Karakoram and associated ranges during cretaceous period.

- The first and oldest phase is formation of Inner Himalayas (formed during Oligocene period).
- The second phase led to formation of Middle Himalayas (Miocene period).
- The third and latest phase was formation of Outer Himalayas (Post Pliocene period).

Outer Himalayas

- Its Vedic name is Siwalik. It is youngest and lowest range of Himalaya.

- The height of this range varies between 600 m to 1500 m, but rarely exceeds 650 m.

- Siwaliks are characterized by fault scraps, anti clinical crests and syn clinical hills.

Middle Himalaya or Lesser Himalaya

- The Middle Himalaya forms the most intricate an rugged mountainous system, 60 – 80 km wide and 3700m to 4500m high.
- Its Vedic name is **Himanchal.**
- Most of the hill stations such as Dalhousie, Manali, Shimla, Nainital, Musoorie, Darjeeling, etc. are located in this range.
- The Kashmir valley which is about 150 km long and 80 km wide lies between the Pir Panjal and the Zaskar ranges

- From west to east middle Himalaya is divided into following ranges:

(i) Pirpanjal range (J&K): It is longest range of the Middle Himalaya.

(ii) Dhauladhar range (Himachal Pradesh)

(iii) Musoorie range (Uttarakhand)

(iv) Nagtiba range (Nepal)

(v) Mahabharat range (Nepal)

Inner Himalaya or Greater Himalaya

- Its Vedic names are Himadri & Bahirgiri. It is the highest mountain range of the world. Its average height is 6100 m.

Important Peaks in Inner Himalaya

- Mt.Everest (8848 m): It is known as Sagarmatha in Nepal and Chomolangma in China. It is the highest peak of the world located in Nepal.

- Mount K-2 (8611 m): It is also known as Godwin Austin or Qagir. It is the 2nd highest peak of the world located in Pak Occupied Kashmir in Karakoram range. Mt. K-2 is highest peak of India.

- Mt.Kunchenjanga (8598 m): It is the third highest peak of the world and 2nd highest of India located in Sikkim.

- Other important peaks: Makalu (8481 m), Nepal: Dhaulagiri (8172 m), Nepal; Manasalu (8156 m), Nepal; Choque (8153 m), Nepal; Nanga Parbat (8108 m), J&K; Kamet and Nandadevi, Uttarakhand, etc.

TRANS HIMALAYA

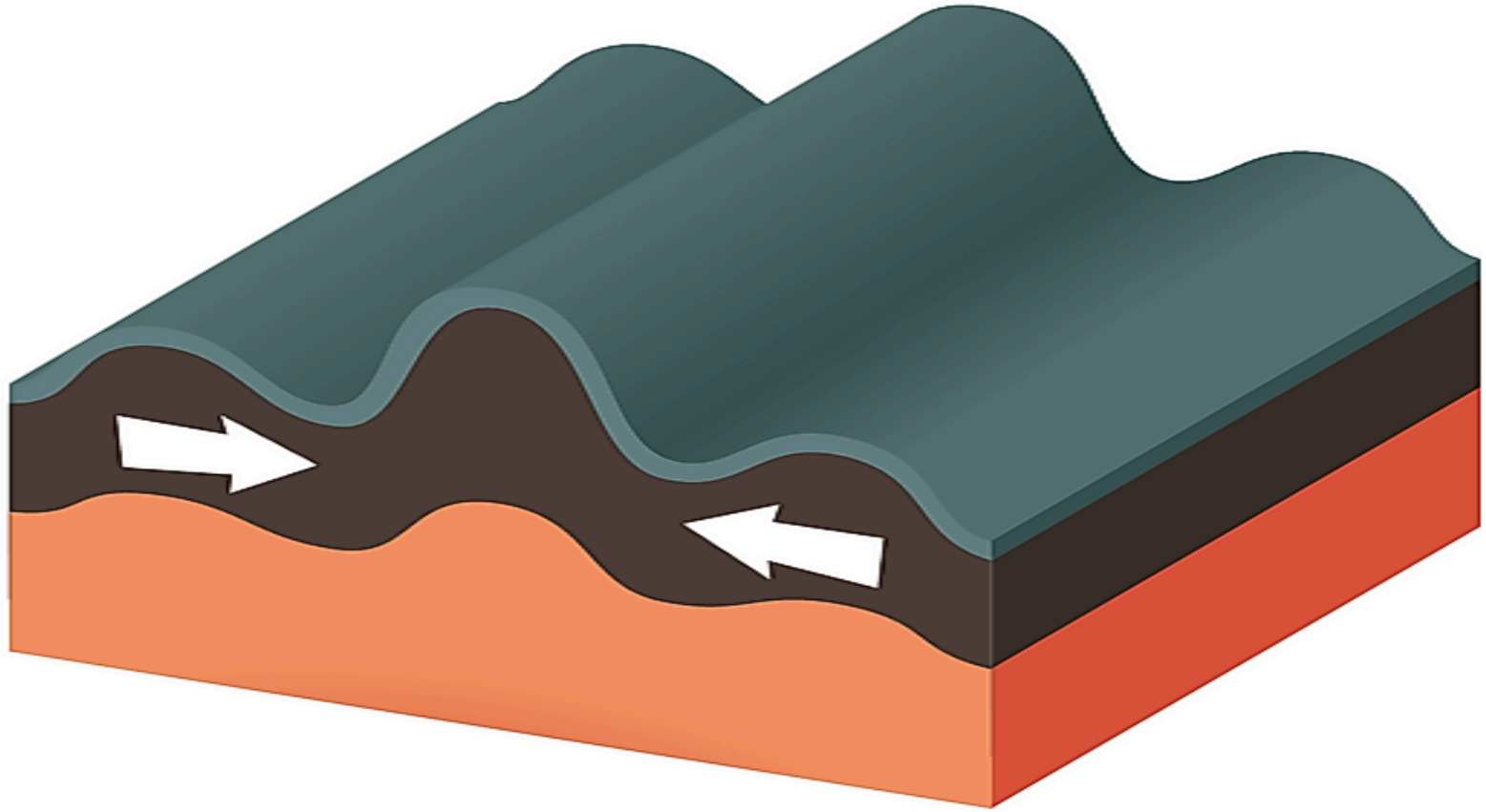
- This region lies behind Himalayas in J&K.
- These are older than other ranges of Himalayas.
- These comprises of Karakoram, Ladakh & Zaskar range. It includes lofty peaks and big glaciers such as Siachin, Baltaro, Biafo and Hispar, etc.

- The highest peak of Karakoram range is Mt. K-2 (8611 m) which is also known as Godwin-Austin or Qagir. The other important peak of this range is Hidden peak (8068 m)

- The highest peak of Ladakh range is Mt. Rakashposhi (7788 m). Gurla Mandhata is another important peak of this range.

- The Kailash range is an offshoot of the Ladakh range. Its average elevation is 5500 to 6000 m. The highest peak of Kailash range is Mt. Kailash (6714 m)

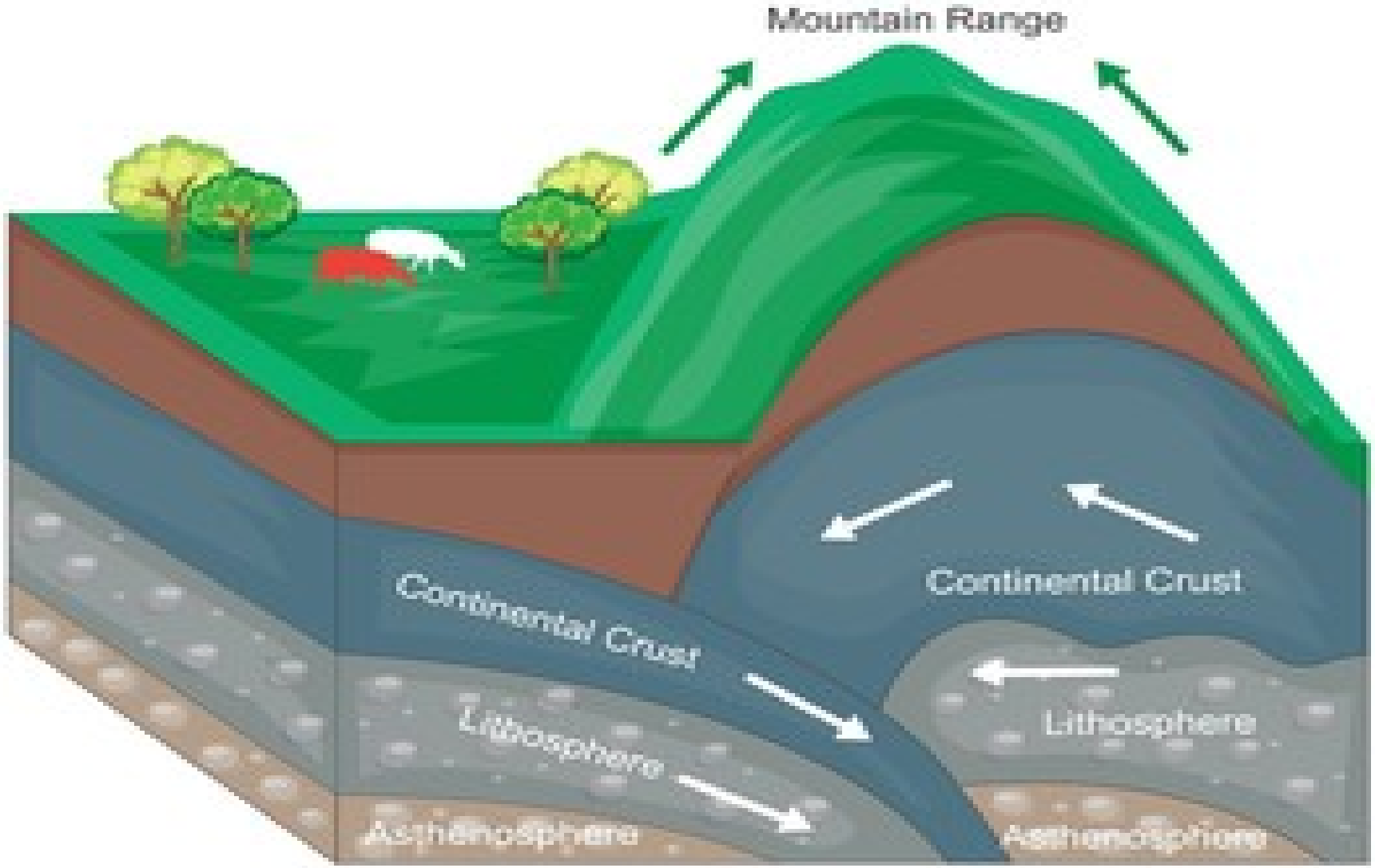
Mountain Building



Folded Mountains

Formed when continental plates collide

MOUNTAIN FORMATION



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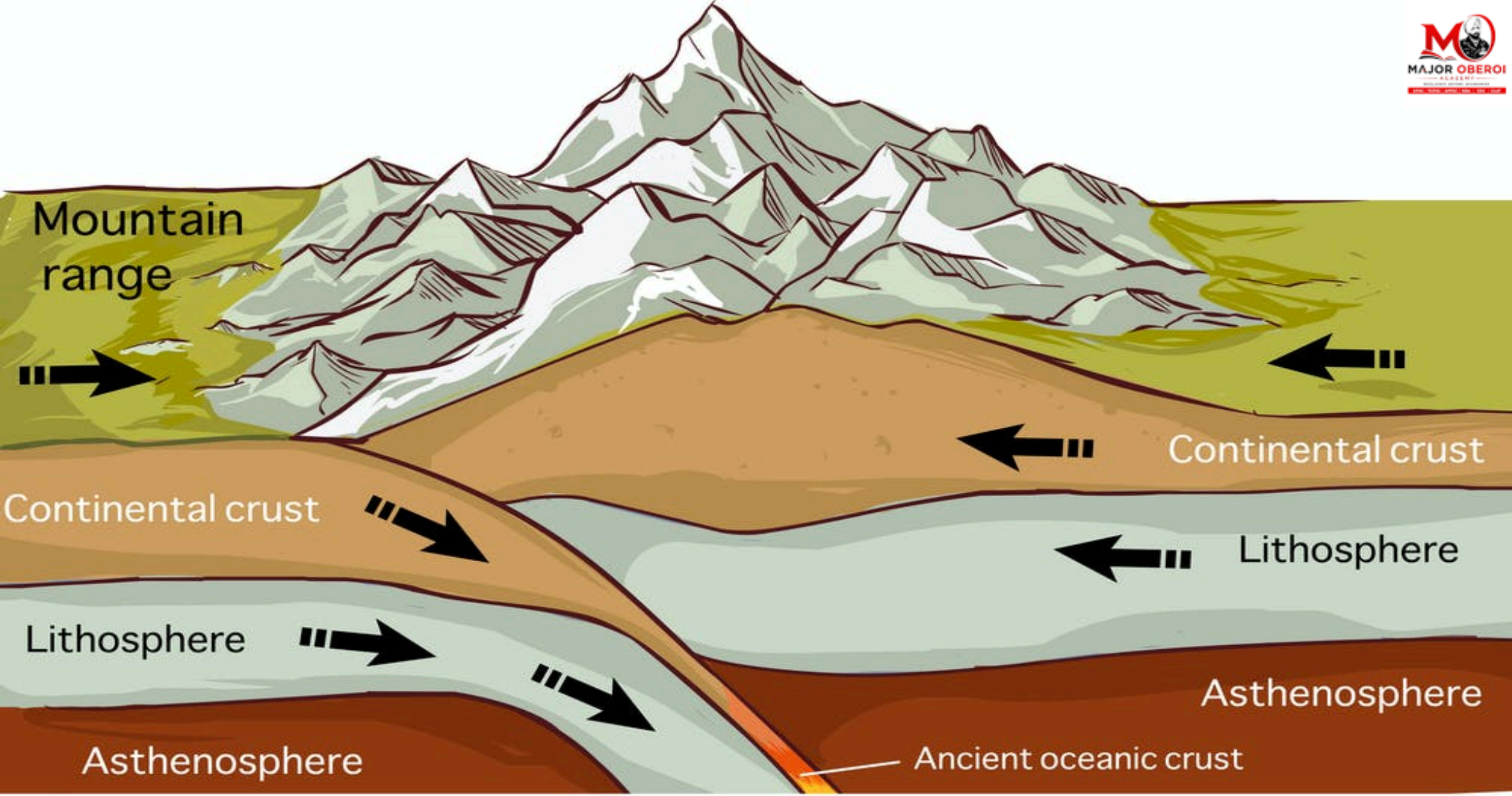
- Orogeny is primarily the mechanism by which mountains are built on continents due to the large structural deformation of the Earth’s lithosphere caused by Plate Tectonics.

- Mountains are formed by movement within the Earth’s crust. The crust itself is made up of several large plates, called tectonic plates, which are free floating. These huge chunks of the Earth’s crust move within molten rock called magma, which allows them to shift and collide over time.

- Even though humans live on the crust, they do not often feel these movements as they are very slow, and the sheer size of each plate is so large.
- Nevertheless, these shifts still have great impacts on human life as the movement of plates is what creates changes in the geographical structure of the surface of the earth. In this way, mountains are created over time.


Mountain range
Continental crust
Continental crust
Lithosphere
Lithosphere
Asthenosphere
Asthenosphere
Ancient oceanic crust

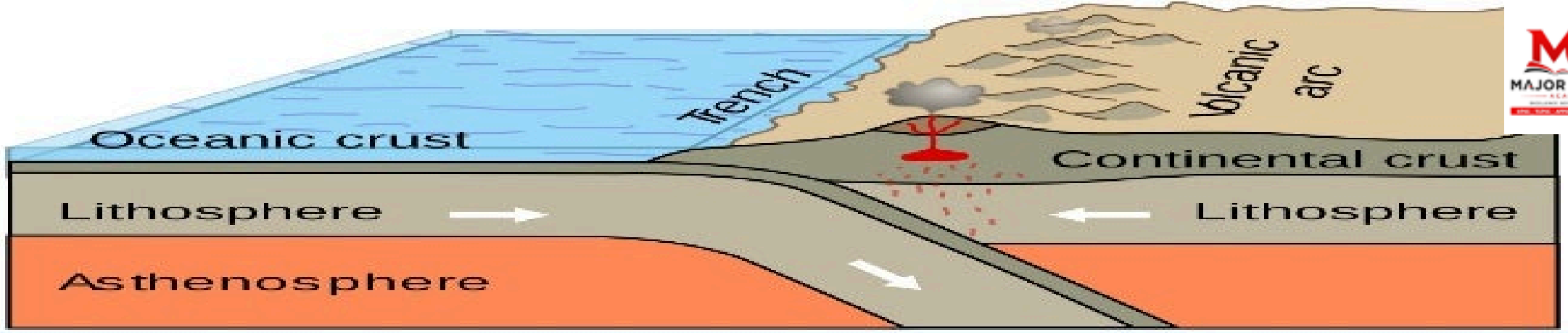
- When these plates collide, there is a great deal of mass and pressure which suddenly comes to a stop, and it is this movement that forces the Earth into buckles or protrusions which are known as mountains.
- Depending on how these plates move or collide, one of three types of mountains can be formed.
- The three types of mountains or mountain ranges are: volcanic, fold, and block mountains, each of which is formed in a different way.

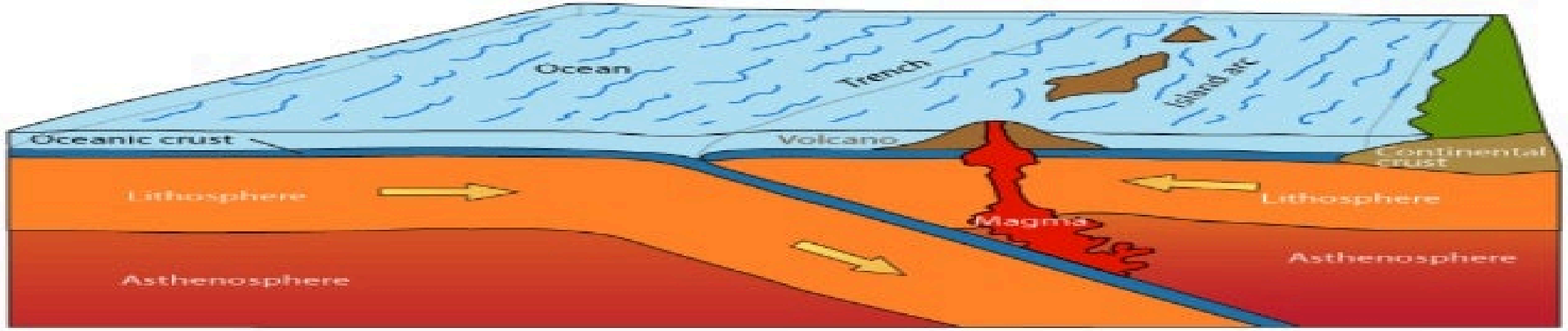
- Mountains are born and have a finite life span. Young mountains are high, steep, and growing upward. Middle-aged mountains are cut by erosion. Old mountains are deeply eroded and often buried.
- We have to note here that the constructive processes, like deformation, folding, faulting, igneous processes and sedimentation build mountains up. On the contrary, the destructive processes like erosion and glaciation, tear them back down again.

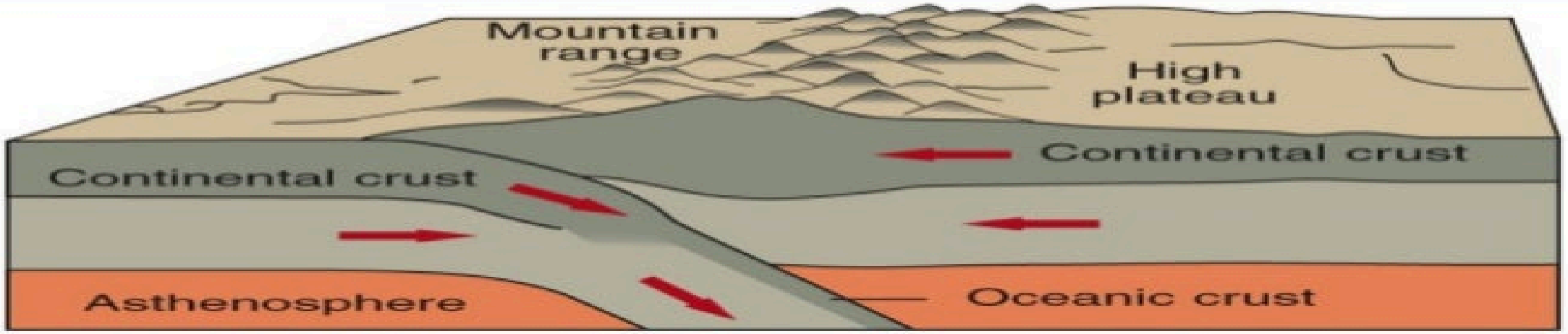
- There are three primary causes of mountain building as follows, which have already studied:
- Convergence at convergent plate boundaries.
- Continental Collisions
- Continent Rifting

- A convergent plate boundary is a location where two tectonic plates are moving toward each other, often causing one plate to slide below the other (in a process known as subduction). The collision of tectonic plates can result in earthquakes, volcanoes, the formation of mountains, and other geological events.

CONVERGENT PLATES

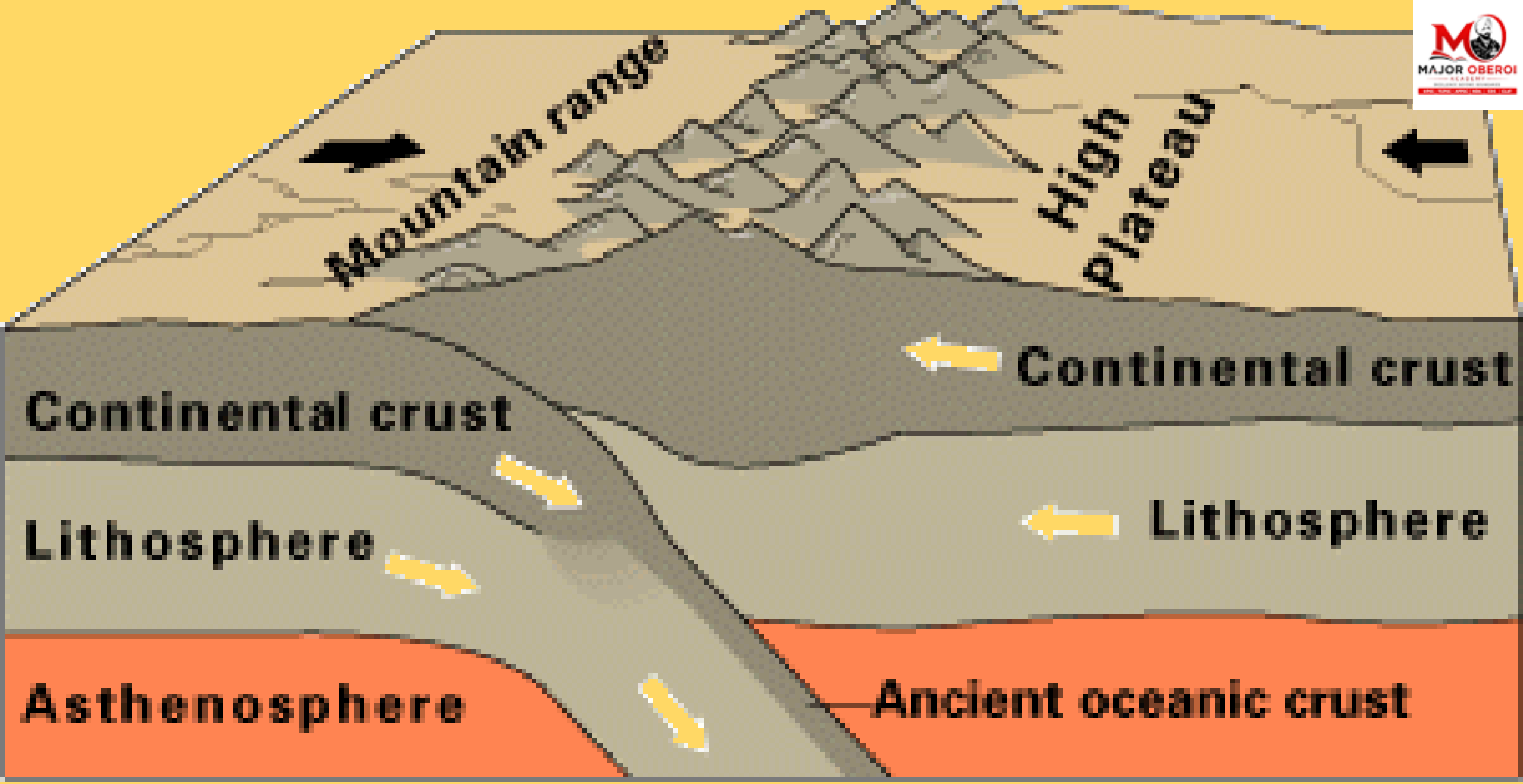






Continental collision is a phenomenon of plate tectonics that occurs at convergent boundaries. Continental collision is a variation on the fundamental process of subduction, whereby the subduction zone is destroyed, mountains produced, and two continents sutured together. Continental collision is only known to occur on Earth.

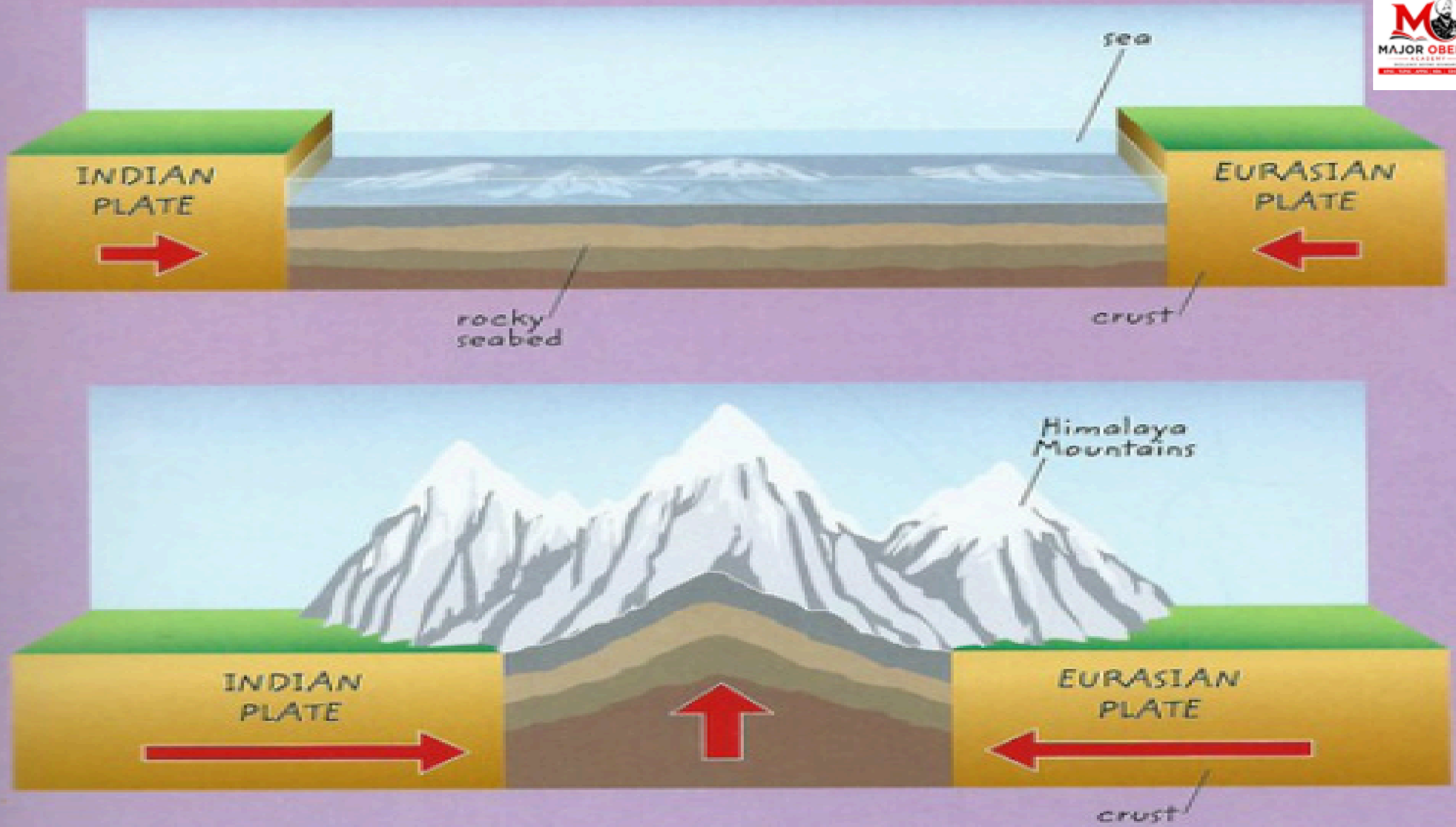
- Continental collision is not an instantaneous event, but may take several tens of millions of years before the faulting and folding caused by collisions stops. The collision between India and Asia has been going on for about 50 million years already and shows no signs of abating. Collision between East and West Gondwana to form the East African Orogen took about 100 million years from beginning (610 Ma) to end (510 Ma). The collision between Gondwana and Laurasia to form Pangea occurred in a relatively brief interval, about 50 million years long.

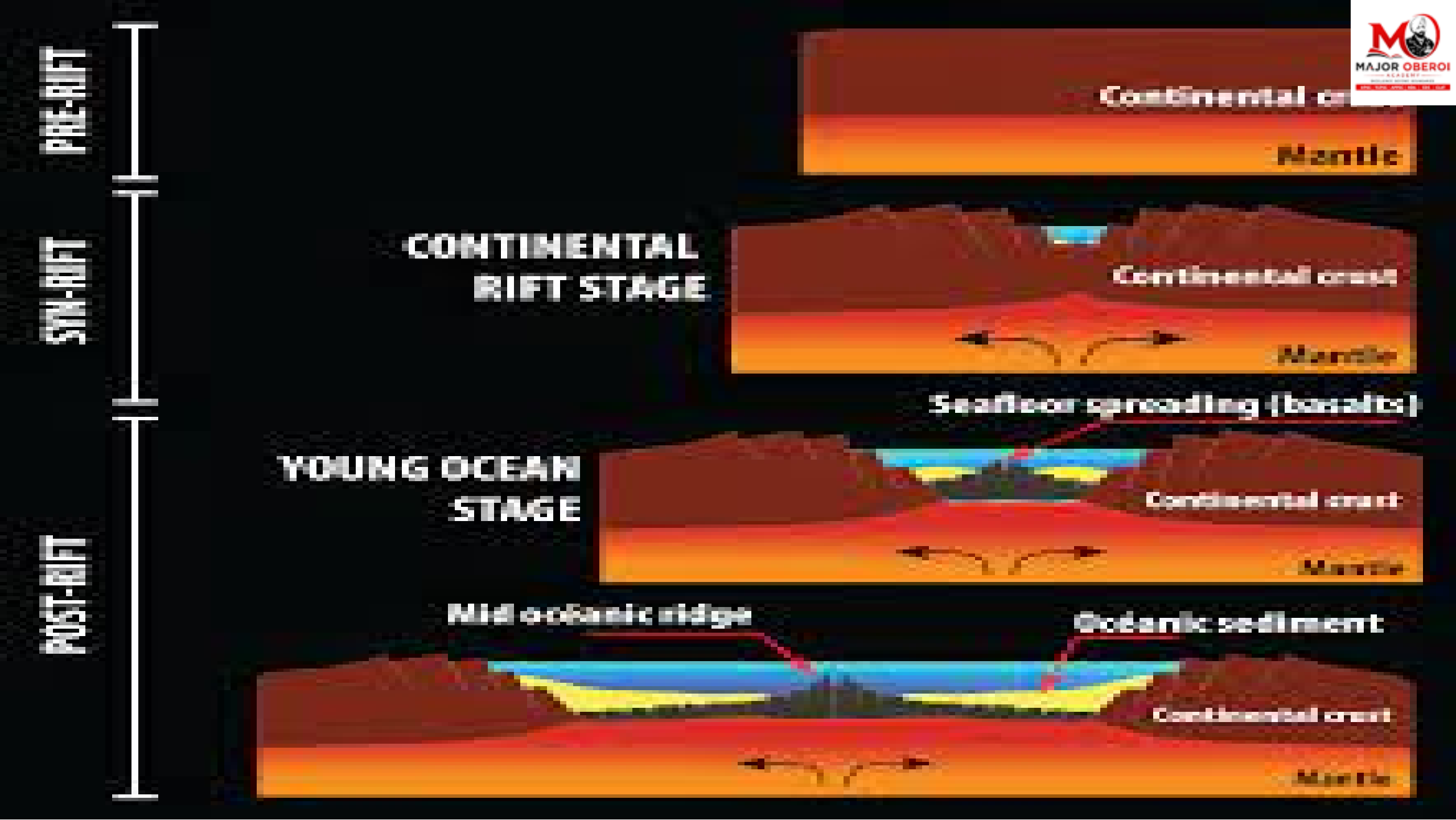


Continental-continental convergence

- A continental rift is conventionally described as a thinning process of the lithosphere ultimately leading to the rupture of the continent and the formation of a mid-oceanic ridge. Rifting is the initial and fundamental process by which the separation of two continents into two tectonic plates takes place

- **“Explain in detail the process of Mountain Formation”.**
- **250 Words**


sea
INDIAN PLATE
EURASIAN PLATE
rocky seabed
crust
Himalaya Mountains
INDIAN PLATE
EURASIAN PLATE
crust


PRE-RIFT
SYN-RIFT
POST-RIFT
CONTINENTAL RIFT STAGE
Continental crust
Mantle
YOUNG OCEAN STAGE
Seafloor spreading (basalts)
Continental crust
Mantle
Mid-oceanic ridge
Oceanic sediment
Continental crust
Mantle

- **Orogenesis involves the following:**
- Structural deformation of the rocks
- Faulting of rocks
- Folding of rocks
- Igneous Processes
- Metamorphism
- Glaciation
- Erosion
- Sedimentation

Structural deformation of the rocks

- When a rock is subjected to increasing stress, it passes through 3 successive stages of deformation.
- Elastic Deformation -- wherein the strain is reversible.
- Ductile Deformation -- wherein the strain is irreversible. Fracture - irreversible strain wherein the material breaks

Faulting of rocks

- A fault is a fracture or zone of fractures between two blocks of rock. Faults allow the blocks to move relative to each other. This movement may occur rapidly, in the form of an earthquake - or may occur slowly, in the form of creep. Faults may range in length from a few millimeters to thousands of kilometers.

Folding of Rocks

- Folds result from the slow deformation of rocks. This happens deep underground where the rocks are under pressure and temperatures are higher. Folded rocks are common in mountain ranges like the Alps, Himalayas and the Scottish Highlands.

Igneous Rock Processes

- Igneous rock is formed when liquid rock freezes into solid rock. This molten material is called magma when it is in the ground and lava when it is on the surface.
- Only the Earth's outer core is liquid; the Earth's mantle and crust are naturally solid.

Metamorphism in Rocks

- Metamorphism is a process of mineral assemblage and texture variation that results from the physical-chemical changes of solid rocks, caused by factors such as crust movement, magma activity, or thermal fluid change in the earth. The metamorphism comprises recrystallization, metamorphic crystallization, deformation, fragmentation, and alternation

Glaciation

- Glaciation is the formation, movement and recession of glaciers. Glaciation was much more extensive in the past, when much of the world was covered in large, continental ice sheets
- Glaciers move by internal deformation of the ice, and by sliding over the rocks and sediments at the base. Internal deformation occurs when the weight and mass of a glacier causes it to spread out due to gravity.

Erosion In Mountain Building

- The tectonic forces that lead to mountain building are continuously countered by erosion due to intensified precipitation, wind and temperature extremes.

Mountain Erosion

The tectonic forces that lead to mountain building are continuously countered by erosion due to intensified precipitation, wind and temperature extremes. These elements, aided by the force of gravity.

Strong winds pick up dust and abrade exposed rock surfaces.

Lightning instantly vaporizes water and ice in rock fissures and literally blows rocks apart.

Temperature changes thaw out and refreeze the ice in rock fissures, wedging them apart, whilst thermal expansion and contraction disintegrates exposed rock surfaces.

Rock falls and ice avalanches scour mountain sides, further eroding the slopes.

- **Forms of Mountains**
- A mountain may have several forms. Important among them are:
- i) mountain ridge, ii) mountain range, iii) mountain chain, iv) mountain system, v) mountain group, and vi) cordillera.

- **Mountain Ridge**
- It is a linear, steep-sided high hill, or spur. The slope of one side of a ridge is steep, while the other side is of moderate slope. A ridge, however, may have symmetrical slopes on both sides.
- The Shimla Ridge is a good example of mountain ridge.

- **Mountain Range**
- A mountain range is a linear system of mountains and hills having several ridges, peaks, summits and valleys.

- **Mountain Chain**
- A mountain chain consists of several parallel long and narrow mountains of different periods.
- **Mountain System**
- A mountain system consists of different mountain ranges of the same period. In a mountain system, different mountain ranges are separated by valleys.
- **Mountain Group**
- A mountain group consists of several unsystematic patterns of different mountain systems

Physiographic divisions of India

- On the basis of physical features, India can be divided into following six divisions
- The Northern mountains
- 2. The Northern Plains
- 3. The Indian Desert
- 4.The Peninsular Plateau
- 5. The Coastal Plains
- 6. The Islands.

The Himalayas
The Great Plains
The Central Highlands
The Peninsular Plateau
The West Coast Plain
The East Coast Plain
The Islands
N
PAKISTAN
CHINA (TIBET)
NEPAL
BHUTAN
MYANMAR
BANGLADESH
ARABIAN SEA
BAY OF BENGAL
Srinagar
Shimla
Chandigarh
Delhi
Jaipur
Lucknow
Patna
Gangtok
Itanagar
Dispur
Shillong
Kohima
Imphal
Aizawl
Agartala
Kolkata
Gandhinagar
Silvassa
Mumbai
Bhubaneshwar
Hyderabad
Panaji
Bangalore
Chennai
Pondicherry (Puduchcheri)
Port Blair
Lakshadweep Islands
Thiruvananthapuram

- The Himalayan Mountains or The Himalayas are the young fold mountains. They run from west-east direction from Indus to Brahmaputra covering a distance of 2500 KM. Their width varies from 400 in the west and 150 KM in the East. The Himalayas may be divided into three parallel ranges:

(a) Greater Himalayas or Himadri

(b) Lesser Himalayas or Himachal

(c) Outer Himalayas or Siwaliks

- **The Greater Himalayas or Himadari**:
- - These are northern most ranges and peaks.
- - average height of 6000 metres and width lies between 120 to 190 Kms .
- - It has high peaks like Mt. Everest, Kanchenjunga, Makalu, Dhaulagiri, Nanga Parbat etc.
- Mt. Everest (8848 m) is the highest peak of the world and Kanchenjunga is the highest peak of Himalaya in India.
- -The passes exist in this range, namely, Bara Lacha-La, Shipki-La, Nathu-La, Zoji-La etc.
- -

- **The Lesser Himalayas or Himachal:**
- - height 1000 and 4500 metres
- - average width is 50 KM.
- - ranges in this are PirPanjal, DhaulaDhar and Mahabharata ranges.
- - hill stations like Shimla, Dalhousie Darjeeling, Chakrata, Mussoorie, Nanital etc.
- - valleys like Kashmir, Kullu, Kangra etc.

- **The Outer Himalayas or the Siwaliks:**
- outer most range of the Himalayas.
- - height varies from 900-1100 meters and the width lies between 10-50 KM.
- - valleys lying between Siwalik and Lesser Himalayas (Himachal) are called ‘Duns’ like DehraDun, Kotli Dun and Patli Dun.

- **The Trans-Himalayan ranges**:
- It extends north of greater Himalaya and parallel to it is called Zaskar range.
- North of Zaskar range lies Ladakh range. The Indus river flows between Zaskar and Ladakh range.
- The Karakoram range lie extreme north of the country. K2 is the second highest peak of the world.

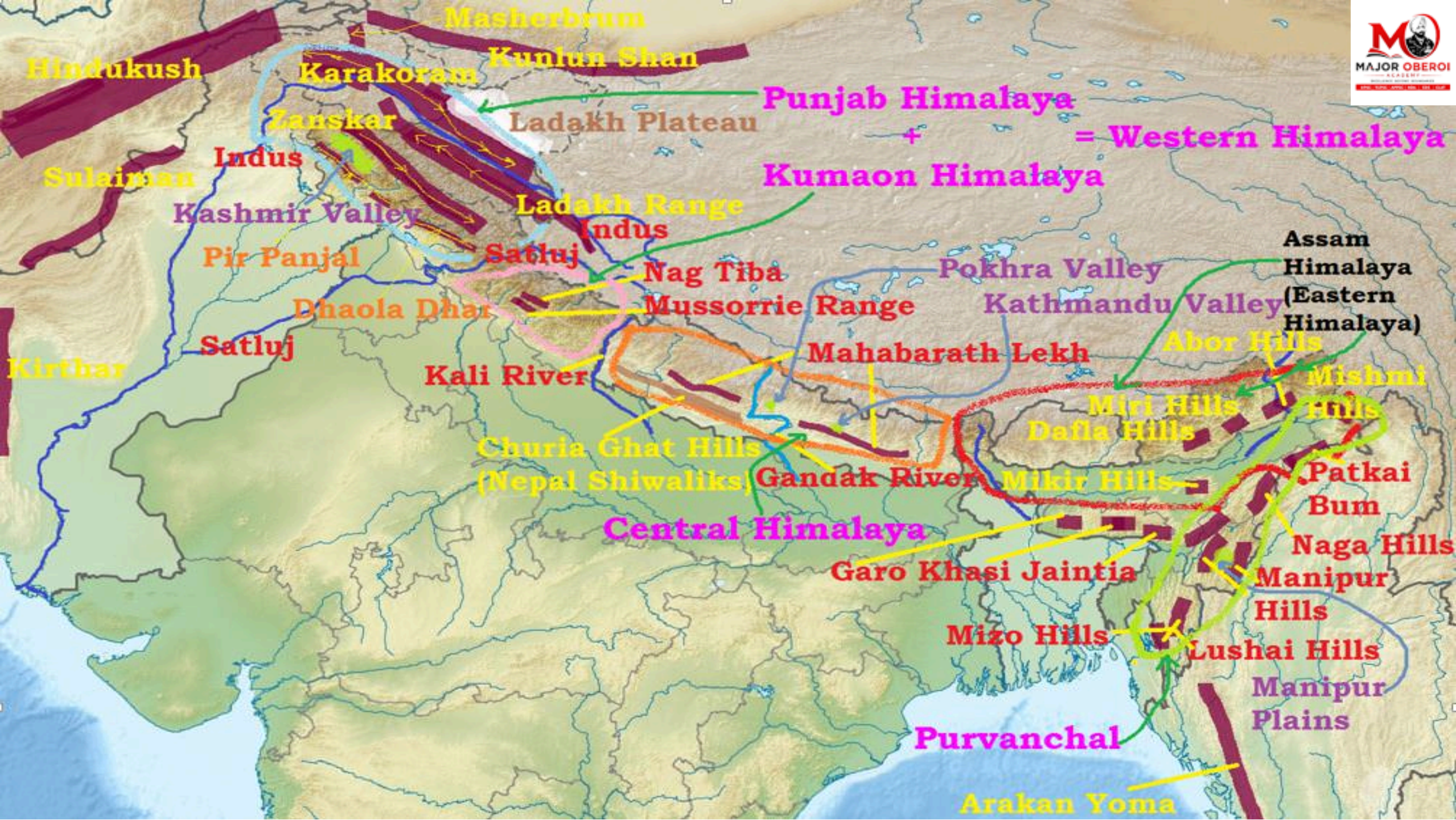
- **The Purvanchal hills**:
- It comprises Mishami, Patkoi, Naga, Mizo hills which are located in eastern side. The Meghalaya plateau is also part of these hills which includes the hills of Garo, Khasi and Jaintia.

Divisions of Himalayas

- There are four main divisions of Himalayan Mountain Ranges which are separated from one another by the gorges of rivers which pass through them.

- **The Punjab Himalayas:**
- The section between the Indus and the Sutlej is known as the Punjab Himalayas. It is also called Jammu-Kashmir and Himachal Himalaya because most of this section is located in these two states. This section is 560 kilometers long. The important ranges of this section are the Ladakh, the PirPanjal, the Dhauladhar and the Zaskar. The Zoji La pass is at an altitude of 3,444 meter above the sea-level.

- Western Himalaya refers to the western half of the Himalayas, stretching from Badakhshan in north eastern Afghanistan/southern Tajikistan, Pakistan through North India Jammu and Kashmir, Ladakh and Himachal Pradesh.
- It is also known as Punjab Himalaya.


Masherbrum
Kunlun Shan
Hindukush
Karakoram
Punjab Himalaya
+
Kumaon Himalaya
= Western Himalaya
Zanskar
Ladakh Plateau
Indus
Sulaiman
Kashmir Valley
Ladakh Range
Indus
Pir Panjal
Satluj
Nag Tiba
Mussorrie Range
Pokhra Valley
Kathmandu Valley
Assam
Himalaya
(Eastern
Himalaya)
Dhaola Dhar
Satluj
Kirthar
Kali River
Mahabarath Lekh
Abor Hills
Mishmi
Hills
Miri Hills
Dafla Hills
Churia Ghat Hills
(Nepal Shiwaliks)
Gandak River
Mikir Hills
Patkai
Bum
Central Himalaya
Naga Hills
Garo Khasi Jaintia
Manipur
Hills
Mizo Hills
Lushai Hills
Manipur
Plains
Purvanchal
Arakan Yoma

- **The Kumaon Himalayas**:
- This section runs for a distance of 320 kilometers between the Sutlej and the Kali rivers. This is higher than the Punjab Himalayas. Great rivers like Ganga and Yamuna have sources in the Kumaon Himalayas.

- **The Nepal Himalayas:**
- This is situated between the Kali and the Teesta rivers and is 800 kilometers long. Most of it is located in Nepal and hence its name. This is the highest part of the Himalayas where lofty peaks like the Everest, the Kanchenjungha, the Dhaulagiri, the Annapurna, and the Makalu etc. are located. The flat Kathmandu valley is also situated in the Nepal Himalayas.

- The Assam Himalaya:
- It extends for a distance of 720 kilometers between the river Tista and the Dihang (Tsangpo-Brahmaputra).